

Post-Quantum Cryptography on Soteria Network: A Comprehensive Trade-off Analysis

A Technical Assessment of ML-DSA and ML-KEM Implementation

Executive Summary

Soteria Network currently operates with:

Metric	Current Value
Block Time	10–12 seconds
Orphan Rate	0.6–0.8%
Node Operating Cost	~\$12/month (3 nodes)
Transaction Size	~110 bytes (ECDSA secp256k1)
User Experience	Near-instant confirmations

This document analyzes the real-world implications of migrating from ECDSA to NIST-standardized Post-Quantum Cryptography (PQC) — specifically ML-DSA for digital signatures and ML-KEM for encrypted P2P

transport.

Part I: Understanding ML-DSA (Module-Lattice-Based Digital Signature Algorithm)

ML-DSA is NIST's standardized post-quantum signature scheme (FIPS 204), derived from the CRYSTALS-Dilithium algorithm. It comes in three security parameter sets.

ML-DSA Parameter Sets Comparison

Parameter Set	NIST Level	Security Equivalent	Public Key	Secret Key	Signature
ML-DSA-44	Level 2	~AES-128	1,312 bytes	2,560 bytes	2,420 bytes
ML-DSA-65	Level 3	~AES-192	1,952 bytes	4,032 bytes	3,309 bytes
ML-DSA-87	Level 5	~AES-256	2,592 bytes	4,896 bytes	4,627 bytes

Source: NIST FIPS 204 specifications

Each level provides progressively stronger quantum resistance. ML-DSA-44 is the most practical for blockchain applications today.

How ML-DSA Compares to ECDSA

Component	ECDSA (secp256k1)	ML-DSA-44	ML-DSA-65	ML-DSA-87
Signature Size	64–65 bytes	2,420 bytes (37x larger)	3,309 bytes (51x)	4,627 bytes (71x)
Public Key Size	32–33 bytes	1,312 bytes (40x larger)	1,952 bytes (59x)	2,592 bytes (79x)
Transaction Size	~110 bytes	~2.5 KB (22x larger)	~3.3 KB (30x)	~4.6 KB (42x)

Source: Testnet Chain post-quantum migration test data

Part II: Understanding ML-KEM (Module-Lattice-Based Key Encapsulation Mechanism)

ML-KEM is NIST's standardized post-quantum key encapsulation mechanism (FIPS 203), designed to establish shared secrets over public channels. It would be required for encrypting Soteria's P2P transport layer.

ML-KEM Parameter Sets

Parameter Set	NIST Level	Public Key	Secret Key	Ciphertext	Shared Secret
ML-KEM-512	Level 1	800 bytes	1,632 bytes	768 bytes	32 bytes
ML-KEM-768	Level 3	1,184 bytes	2,400 bytes	1,088 bytes	32 bytes
ML-KEM-1024	Level 5	1,568 bytes	3,168 bytes	1,568 bytes	32 bytes

Source: NIST FIPS 203 specifications

ML-KEM-768 is recommended as the balanced choice for most applications.

Part III: Real-World Performance Impact — The Testnet Chain Case Study

Soteria Chain conducted the real-world test of post-quantum cryptography on a testnet blockchain. Their findings are directly relevant to Soteria Network.

Before and After: Test Results

Metric	Before (ECDSA)	After (ML-DSA-44)	Change
Transaction Signature	65 bytes	2,420 bytes	+3,623%
Full Transaction Size	~110 bytes	~2.5 KB	+2,173%
Block Size (at 2,000 TPS)	~130 KB	~2 MB	+1,438%

Sources: Testnet Chain Post-Quantum Cryptography

Key Insights from the Test

The Bottleneck Is Data Propagation, Not Computation: ML-DSA verification is actually faster than ECDSA — ML-DSA-65 achieves verification in ~0.14ms compared to 0.88ms for ECDSA. The throughput drop comes

entirely from larger blocks taking longer to propagate across the network.

Cross-Region Performance Suffers Most: The finality latency increase was "driven entirely by block propagation delays over wider network links, not by any issue within the consensus protocol itself".

Consensus Aggregation Can Be Optimized: pqSTARK aggregation can compress signatures into a 43:1 compression ratio.

Compatibility Preserved: Wallet addresses remain unchanged (20 bytes, derived from ML-DSA-44 public key), so existing wallets, and RPCs require no redesign.

Part IV: Projected Impact on Soteria Network

Applying Testnet Chain's findings to Soteria's current metrics:

Before PQC Implementation (Current State)

Metric	Current Value	Notes
Block Time	10–12 seconds	Hardcoded 12s, ~10s with many miners
Orphan Rate	0.6–0.8%	Good network stability
Transaction Size	~110 bytes	ECDSA secp256k1
Node Count	Multiple	Decentralized
Node Cost	Affordable	Sustainable long-term

P2P Encryption	Classical ECDH	Vulnerable to future quantum attack
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After ML-DSA-44 Implementation (Projected)

Metric	Projected Value	Change
Block Time	30–60 seconds (required)	+150–400%
Orphan Rate	3–5%+ (estimated)	+400–700%
Transaction Size	~2.5 KB	+2,173%
Node Storage Requirements	5–10x current	+400–900%
Node Cost (3 nodes)	\$60–\$120/month	+400–900%
P2P Encryption	ML-KEM-768 (1088-byte ciphertexts)	New overhead
User Experience	Significant delays	Loss of "near-instant" feel

After ML-DSA-65 or ML-DSA-87 Implementation (Projected)

Metric	ML-DSA-65	ML-DSA-87
Transaction Size	~3.3 KB	~4.6 KB
Projected TPS Drop	~50–55%	~60–65%
Block Time Required	45–75 seconds	60–90 seconds

Orphan Rate Risk	High	Very High
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Part V: The Cost-Benefit Analysis

What You Gain

- **Quantum Resistance:** Protection against future quantum computers capable of running Shor's algorithm
- **NIST Standardization:** Algorithm has been vetted by global cryptographic community
- **Future-Proofing:** No urgent migration needed later

What You Lose

- **Block Time:** From 10–12 seconds to 30–60+ seconds — the "instant" experience disappears
- **Orphan Rate:** From 0.6–0.8% to potentially 3–5%+ — network less stable
- **Node Costs:** From \$12/month to \$60–\$120+/month — centralization risk increases
- **Throughput:** 40–50% drop in transaction capacity
- **P2P Overhead:** ML-KEM adds 1,088-byte ciphertexts to every encrypted handshake
- **Development Complexity:** Significant code changes, testing, and audit costs

Critical Consideration: The Quantum Timeline

The reports explicitly noted that "quantum computers are **'not yet at a stage'** where they can break current production cryptography in real-world systems". Researchers concluded that "cryptographically relevant quantum computers remain **at least a decade away**".

This means implementing PQC today is a **defensive downgrade** — you are sacrificing immediate performance, user experience, and decentralization for protection against a threat that does not yet exist.

Part VI: The Orphan Rate Risk — A Detailed Analysis

Orphan rates are particularly sensitive to block propagation delay. With ML-DSA:

- **Block sizes increase by ~15x** (from ~130 KB to ~2 MB under similar transaction loads)
- **Larger blocks take longer to propagate** across geographically distributed nodes
- **Longer propagation = higher orphan probability** — two miners finding blocks simultaneously becomes more likely
- **Testnet Chain's finality latency jumped into a 450% increase**

For Soteria's 10–12 second block time, this is catastrophic. Even with a conservative estimate, the orphan rate would likely climb from **0.6–0.8% to 3–5%+**, significantly reducing network security.

Part VII: The ML-KEM Question for P2P Transport

If Soteria implements quantum-resistant P2P encryption:

Component	Current (ECDH)	With ML-KEM-768
Handshake Size	~64 bytes	~1,088 bytes ciphertext

Key Exchange Speed	Very fast	Slower (lattice operations)
Node CPU Overhead	Minimal	Increased (key generation, encapsulation)

This adds another layer of overhead — every peer-to-peer connection establishment becomes heavier and slower.

Part VIII: Recommendations for Soteria Network

Short-Term (0–2 Years): Do Not Implement PQC

1. **Preserve your competitive advantage:** 10-second blocks, 0.8% orphans, \$12/month nodes (dev and admins nodes) are exceptional metrics
2. **Let larger chains test first:** Bitcoin, Ravencoin, Ethereum have far more resources to handle the migration pain
3. **Monitor Bitcoin & Ravencoin Chain's progress:** They are actively working on scaling solutions
4. **Focus on what makes Soteria unique:** Fast finality, low node costs and migration to a better mining algorithm

Medium-Term (2–5 Years): Prepare, Don't Deploy

1. **Research pqSTARK aggregation:** Chain achieved 43:1 compression for consensus votes
2. **Evaluate hybrid approaches:** Allow both ECDSA and ML-DSA signatures during transition
3. **Monitor quantum computing advances:** The threat may arrive faster or slower than predicted

Long-Term (5–10+ Years): Deploy When Necessary

1. **Implement ML-DSA-44** (smallest footprint, sufficient security margin)
2. **Consider increasing block time** to 30–60 seconds to accommodate larger blocks
3. **Optimize aggressively:** Use pqSTARK or similar aggregation for consensus

Part IX: Summary Table — Before vs. After PQC Implementation

Metric	Current (ECDSA)	ML-DSA-44	ML-DSA-65	ML-DSA-87
Block Time	10–12 sec	30–60 sec	45–75 sec	60–90 sec
Orphan Rate	0.6–0.8%	3–5%+	5–8%+	8–12%+
Transaction Size	~110 B	~2.5 KB	~3.3 KB	~4.6 KB
TPS (relative)	100%	~60%	~50%	~40%
Block Size	Small	~2 MB	~2.7 MB	~3.8 MB
Node Cost (3 nodes)	~\$12/mo	~\$60–120/mo	~\$80–160/mo	~\$100–200/mo
P2P Overhead	Low	+1,088 B/ciphertext	Same	Same
Quantum Resistance	✗ None	✓ Level 2	✓ Level 3	✓ Level 5
User Experience	★ Excellent	★★ Good	★ Acceptable	★ Poor

Part X: Conclusion

Post-Quantum Cryptography is **real, necessary, and coming**. However, the data from Testnet Chain's production test shows that implementing it today comes with **severe and immediate trade-offs**:

- **40–50% throughput drop**
- **22x larger transactions**
- **450% increase in worst-case finality latency**
- **5–10x higher node operating costs**
- **Significantly higher orphan rates**

For Soteria Network, which has built its identity around **speed (10-12 second blocks), efficiency (low node costs), and user experience (near-instant confirmations)**, implementing PQC today would **destroy the very advantages that make the blockchain attractive**.

The quantum threat is real but **not imminent**. The rational approach is to **wait, monitor, and prepare** — letting larger ecosystems like Bitcoin, Ethereum and Ravencoin absorb the initial migration costs and optimization lessons.

When the time comes (5–10 years from now), the technology will be more mature, the implementations will be more efficient, and the migration will be less painful. Until then, Soteria Network should **stay fast, and stay cheap**.
